

The active integration of information during learning with dynamic and interactive visualisations

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Abstract

Computer-based learning environments commonly comprise symbolic as well as static and dynamic pictorial representations, frequently combined with the possibility of modifying them interactively. While multiple, dynamic and interactive external representations have the potential to improve learning in various ways, they also place specific demands on learners, such as the need to process and relate different representations, to control and evaluate interactions with these representations, and to construct coherent mental representations. Because learners frequently are not able to meet these demands, the presentation of multiple, dynamic and interactive representations might not only not improve but might even impede learning. Starting from cognitive load theory as well as from structure mapping theory, we developed support measures that encouraged learners to actively integrate symbolic and pictorial representations and to interact with dynamic pictorial representations in a structured and reflective way. In two experimental studies, the learning effects of the encouraged activities were evaluated. Analyses of variance revealed (1) that the active integration of different representations improved learning significantly and (2) that the structured interaction with different representations specifically increased verbal understanding.

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Frequently, it is taken for granted that the use of dynamic visualisations in computer-based learning environments is beneficial to learning. However, dynamic visualisations not only offer varied potential to learners but also place specific requirements on learners. These requirements are related to the dynamics and the interactivity of the visualisations, as well as to the features of the learning environment the visualisations are embedded in.

For example, dynamic visualisations may improve learning by illustrating abstract time-dependent concepts, by simulating complex system behaviour, by demonstrating sequential processes, and by focussing and guiding the learners' attention (Park & Hopkins, 1993; Rieber, 1990). However, learners have to process large amounts of information displayed in different regions of the screen and thus have to direct their attention simultaneously to different sources of information. In addition, because the displayed information may change continuously, it can only be viewed for a limited amount of time (e.g., Lowe, 1998, 1999). Very often, these requirements overburden the learners' cognitive capabilities resulting in inadequate learning. Furthermore, the externalisation of dynamic processes may even prevent learners from performing cognitive processes relevant to learning (e.g., Schnotz, Boeckheler, & Grzondziel, 1999). In order to cope with these requirements, learners frequently make use of a strategy that limits their processing to selected aspects of a dynamic visualisation. Very often, however, these aspects are not the most relevant of the visualisation, but rather those that are most perceptually compelling (Lowe, 2003).

The use of interactive visualisations frequently aims at inducing active learner behaviour and constructive learning processes (e.g., Schnotz, 2002; Schnotz & Banert, 1999). For example, in simulation environments for discovery learning, learners attempt to infer the structure of an underlying conceptual model by changing the values of input variables in a static or dynamic way and by observing the consequences of these changes in the presented symbolic and pictorial representations. However, many learners do not interact with the presented external representations in a structured and goal-oriented way and fail to formulate, test, and evaluate hypotheses systematically (de Jong & van Joolingen, 1998; Reigeluth & Schwartz, 1989).

Usually, dynamic and interactive visualisations are presented in combination with other kinds of external representations such as static pictures, text, and formulas. These different representations may complement each other, resulting in a more complete representation of an application domain (e.g., Ainsworth, 1999). However, learners are frequently not able to systematically relate multiple external representations to each other. As a consequence, these learners fail to integrate the different sources of information into coherent mental representations, resulting in fragmentary and disjointed knowledge structures (e.g., Ainsworth, Bibby, & Wood, 2002). During problem solving, for example, these learners might switch back and forth between different mental representations of a posed problem, without being able to determine which representation contributes in meaningful ways to the problem's solution (e.g., Anzai, 1991).

In order to enable learners to take advantage of the potential of dynamic and interactive visualisations in computer-based learning environments, they need

support in coping with the specific requirements of these visualisations (e.g., Lewalter, 2003; Lowe, 2003) and in relating them systematically to other external representations (e.g., Gutwill, Frederiksen, & White, 1999; Seufert, 2003).

1. Supporting learning with dynamic and interactive visualisations

Various guidelines to assist in the presentation of information in order to increase learning performances are provided by cognitive load theory (e.g., Chandler & Sweller, 1991; Sweller, 1988; Sweller & Chandler, 1991; Sweller, van Merriënboer, & Paas, 1998). Based on the assumptions of (1) an unlimited long-term memory and (2) a limited working memory (e.g., Baddeley, 1986), cognitive load theory aims at designing instructions that do not overburden the learners' working memory capacities. Three sources of cognitive load are distinguished within cognitive load theory: (1) intrinsic cognitive load, which is determined by the complexity of the application domain as well as by the learners' prior knowledge, (2) extraneous cognitive load, which corresponds to the mental effort imposed by the way information is presented externally, and (3) germane cognitive load, which is a result of mental activities that are directly relevant to learning.

For many years, research on cognitive load theory focused on instructional design intended to decrease extraneous cognitive load. More recently, various studies have been conducted that focus on increasing germane cognitive load. These studies aim at further improving instruction in a way that supports the use of free working memory capacity during learning (Kirschner, 2002; Sweller et al., 1998; van Merriënboer, Schuurman, de Croock, & Paas, 2002).

Following the direction of these research activities, our aim has been to improve learning with dynamic and interactive visualisations in computer-based learning environments using approaches intended to both reduce extraneous cognitive load and increase germane cognitive load. Therefore, we developed our own support measures and re-interpreted several support measures that are already suggested in the literature.

For example, to support learners in dealing with the complexity of dynamic visualisations, it has been suggested that both visual and temporal cues be provided to the learners (e.g., Lowe, 1999), that the underlying conceptual model be presented to the learners step by step (e.g., Swaak, van Joolingen, & de Jong, 1998), and that learners be given the opportunity to process static information before viewing dynamic visualisations (e.g., Mayer & Chandler, 2001). In order to support learning with interactive simulations, it has been repeatedly suggested to structure the learners' interactions with the learning environment (e.g., Spada, Reimann, & Häusler, 1983; van Joolingen & de Jong, 1991). These support measures typically guide learners to (1) focus on specific variables of the underlying model, (2) generate hypotheses about relationships between these variables, (3) conduct simulated experiments in order to test the hypotheses, and (4) evaluate the hypotheses in light of the observed results. It has been demonstrated, however, that successful learning with interactive simulations requires further support. In particular, generating and testing hypotheses seem to be very demanding tasks. De Jong

and his colleagues have proposed several measures to support learners in carrying out these tasks, such as to guide the formulation of specific hypotheses as well as the design of appropriate simulated experiments, or to provide learners with assignments or guiding questions (e.g., [Njoo & de Jong, 1993](#); [Swaak et al., 1998](#)).

Linking these recommendations to cognitive load theory, we suggest that some of the suggested support measures be combined in a way that reduces extraneous cognitive load and increases germane cognitive load.

On the one hand, we have tried to increase germane load by encouraging the learners to formulate, test, and evaluate hypotheses in a structured and reflective way. On the other hand, we have tried to decrease extraneous load by presenting static versions of the dynamic visualisations to the learners before dynamic versions are presented to them. The initial presentation of static visualisations gives learners time to identify the presented visual structures and become familiar with them. Furthermore, as will be described in more detail below, the initial presentation of static visualisations allows learners to relate the possibly unfamiliar visual structures to representations already familiar to them. In addition, both the hypothesis space and the experiment space (cf. [Klahr & Dunbar, 1988](#)) are reduced by encouraging the learners to formulate hypotheses on only one relevant aspect of the visualisation at a time and by providing them examples of data that have the potential to highlight important relationships.

As mentioned before, learners not only need support in coping with the specific requirements of dynamic and interactive visualisations, they also need further support in relating the different presented sources of information to each other.

For example, John Sweller and his colleagues demonstrated in a series of experiments that the spatial integration of textual and pictorial information can reduce unnecessary visual search resulting in a decrease of extraneous cognitive load and better learning. (e.g., [Chandler & Sweller, 1991, 1992](#); [Tarmizi & Sweller, 1988](#)). This so-called split-attention effect has been repeatedly confirmed by Mayer and is the origin of his spatial contiguity principle ([Mayer, 1997, 2001](#)). Another suggested measure to support learners in making connections between different sources of information is to link the features of multiple representations by various symbolic conventions such as the use of the same colour for corresponding entities in different representations (e.g., [Kalyuga, Chandler, & Sweller, 1999](#); [Kozma, 2003](#); [Kozma, Russell, Jones, Marx, & Davis, 1996](#)).

While these measures have the potential to reduce extraneous cognitive load, they do not directly support learners in constructing meaningful knowledge. While viewing integrated sources of information, learners may nevertheless remain rather passive, still unable to mentally process and integrate the represented information in an adequate way. According to the theories of [Mayer \(1997, 2001\)](#) as well as [Schnotz & Bannert \(1999, 2003\)](#), however, the integration of textual and pictorial information into coherent mental representations is essential for successful learning.

[Schnotz, Picard and Hron \(1993\)](#) made use of [Gentner's \(1983\)](#) structure-mapping theory to describe the process of relating graphics and mental representations during learning. According to structure-mapping theory, an analogy between a familiar and an unfamiliar domain is constructed by (partially) mapping the

mental structure that represents the familiar domain onto the mental structure that represents the unfamiliar domain (Gentner, 1983; Gentner & Markman, 1997).

In our research, we take advantage of the concept of structure-mapping in order to support the mental integration of different sources of information by encouraging learners (1) to identify relevant structures in the representations and (2) to spatially integrate different sources of information by systematically and interactively relating familiar representations to unfamiliar representations in the external environment.

We hypothesize that the (inter-) active integration of different sources of information in the external environment supports learning by inducing germane cognitive load. At the same time, the spatial integration of different sources of information gradually leads to an integrated format and—in the end—to a reduction of extraneous cognitive load.

We tested our hypothesis in two experiments. In the first experiment, we made use of learning materials that consisted of static symbolic and pictorial representations. In the second experiment, static symbolic and dynamic pictorial representations were employed. However, before the learners took advantage of the dynamic pictorial representations, they were presented with static versions of the visualisations.

2. Experiment 1

2.1. Method

2.1.1. Design

In a first experiment, we encouraged learners to actively integrate symbolic and static pictorial representations. Two factors were varied: (1) the format of information presentation (split-source format versus integrated format) and (2) the learners' activities (only mentally relating information versus externally and mentally relating information). Correspondingly, four experimental groups were formed.

In the first two experimental groups, the students were required to mentally relate the presented information. While the students of the first group were presented a split-source format, the students of the second group were presented an integrated format. Thus, these two experimental conditions corresponded to conditions that were investigated by Chandler and Sweller (1991).

In the third and fourth experimental group, the students had to externally and mentally relate the presented information. In both groups, the students started from a split-source format. However, while the students of third experimental group retained the split-source format, the students of the fourth experimental group produced an integrated format step by step.

2.1.2. Participants

Overall, 81 students (58 females and 23 males) in the social sciences at the University of Freiburg, aged 19–32, were randomly assigned to the four experimental groups. They were paid for their participation.

2.1.3. *Materials and procedure*

The materials consisted of the learning materials as well as a learning test. The learning materials were made up of symbolic and static pictorial representations that were presented to the students on a computer. They consisted of illustrations and corresponding captions about how a tire pump works, similar to the illustrations used by Mayer, 1997, 2001). Fig. 1 shows the two screens that were presented to the students who had to externally and mentally relate the presented information. While the students of the third experimental group had to relate the symbolic and pictorial information by interactively dragging and dropping corresponding numbers (cf. Fig. 1a), the students of the fourth experimental group had to relate symbolic and pictorial information by interactively producing an integrated format (cf. Fig. 1b).

In a pre-study it was demonstrated that on the one hand, students of the social sciences know—of course—what a tire pump is, but on the other hand possess almost no adequate knowledge about how a tire pump works. Therefore, only a post-test was administered to the students in this study. The post-test was made up of eight multiple choice questions that assessed the students' retention of information directly given in the learning materials as well as seven multiple choice questions that assessed the students' understanding of the causal mechanisms underlying the information given in the learning materials.

The students worked on the learning materials individually for 4 min. The students who had to externally relate the symbolic and pictorial information were first given the opportunity to practice dragging and dropping information on a computer. Finally, the students answered the multiple choice questions.

2.2. *Results*

Table 1 shows the means and the standard deviations for retention and understanding for all four experimental conditions. Table 2 shows the results of the multivariate and univariate two-way analyses of variance.

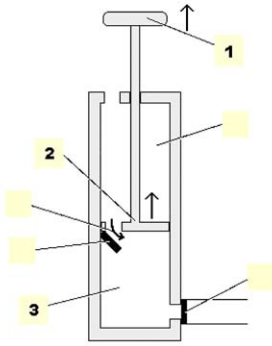
On average, learning with an integrated format was more successful than learning with a split-source format. Statistically, this finding was confirmed with respect to the students' understanding. In addition, externally and mentally relating information was on the average more beneficial to learning than only mentally relating information. However, this result did not reach statistical significance. There was no notable interaction observable between the format of information presentation and the learners' activities.

2.3. *Discussion*

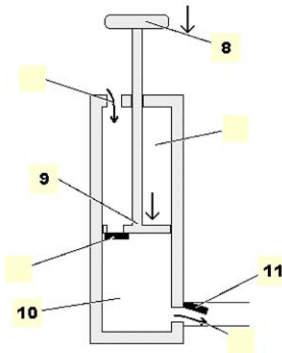
The first experiment yielded two results. Firstly, presenting symbolic and static pictorial representations in an integrated format facilitates learning significantly more than presenting them in a split-source format. This result replicates once more the findings by John Sweller and his colleagues (e.g., Chandler & Sweller, 1991; Sweller & Chandler, 1994). Secondly, learning by actively producing an integrated format tends to be more successful than learning by merely reconstruct-

(a)

- | | |
|--|---|
| 1 Der Griff wird nach oben gezogen. | 5 In diesem Bereich der Luftpumpe herrscht Oberdruck. |
| 2 Der Kolben bewegt sich nach oben. | 6 Das Einlassventil ist offen. |
| 3 In diesem Bereich der Luftpumpe herrscht Unterdruck. | 7 Luft gelangt vom oberen in den unteren Bereich der Luftpumpe. |
| 4 Das Auslassventil ist zu. | |

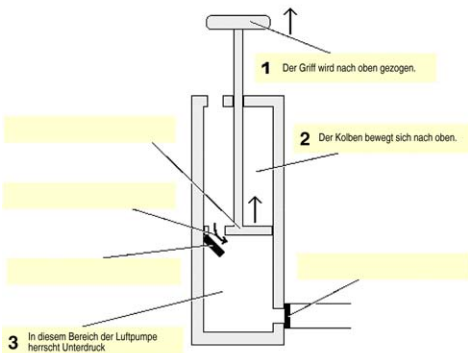


- | | |
|--|---|
| 8 Der Griff wird nach unten gedrückt. | 12 Das Brilassventil ist geschlossen. |
| 9 Der Kolben bewegt sich nach unten. | 13 Luft gelangt durch das Auslassventil in den Schlauch. |
| 10 In diesem Bereich der Luftpumpe herrscht Oberdruck. | 14 In diesem Bereich der Luftpumpe herrscht Unterdruck. |
| 11 Das Auslassventil ist offen. | 15 Durch das obere Loch gelangt Luft in den oberen Bereich der Luftpumpe. |



(b)

- | | |
|-----------------------------|---|
| | 5 In diesem Bereich der Luftpumpe herrscht Oberdruck. |
| | 6 Das Einlassventil ist offen. |
| | 7 Luft gelangt vom oberen in den unteren Bereich der Luftpumpe. |
| 4 Das Auslassventil ist zu. | |



- | | |
|--|---|
| | 12 Das Brilassventil ist geschlossen. |
| | 13 Luft gelangt durch das Auslassventil in den Schlauch. |
| | 14 In diesem Bereich der Luftpumpe herrscht Unterdruck. |
| | 15 Durch das obere Loch gelangt Luft in den oberen Bereich der Luftpumpe. |

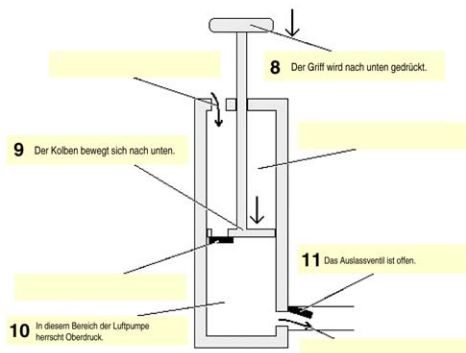


Fig. 1. Relating symbolic and pictorial information (a) by interactively dragging and dropping corresponding numbers and (b) by producing an integrated format.

ing an already integrated format. However, the observed differences in learning gains were very small.

Table 1

The means (M) and the standard deviations (SD) for retention and understanding under all four experimental conditions

Activity		Retention			Understanding		
		Information presentation					
		Split-source	Integrated	Overall	Split-source	Integrated	Overall
Mentally	M	23.50 (73%)	25.75 (81%)	24.63 (77%)	17.05 (61%)	19.00 (68%)	18.03 (64%)
Relating	SD	3.62	4.18	4.02	3.17	3.69	3.53
Externally	M	25.00 (78%)	25.89 (81%)	25.49 (80%)	17.63 (63%)	20.74 (74%)	19.31 (69%)
Relating	SD	3.44	3.26	3.33	3.20	2.40	3.17
Overall	M	24.17 (76%)	25.82 (81%)	25.03 (78%)	17.31 (62%)	19.85 (71%)	18.63 (67%)
	SD	3.57	3.71	3.72	3.15	3.21	3.41

Table 2

The results of the multivariate (Wilks-Lambda) and univariate two-way ANOVA

Source of variance	Dependent variable	df	F
Information presentation	Across retention and understanding	2, 70	6.02**
	Retention	1, 71	3.43
	Understanding	1, 71	11.97**
Activity	Across retention and understanding	2, 70	1.29
	Retention	1, 71	0.94
	Understanding	1, 71	2.50
Information presentation × activity	Across retention and understanding	2, 70	1.06
	Retention	1, 71	0.64
	Understanding	1, 71	0.63

** $p < 0.01$.

This is probably due to the fact that the difficulty of the learning task was comparatively low. The average relative solution frequencies in the post-test (cf. Table 1) suggest that retaining and understanding the presented materials did not really demand extensive elaboration. While the learning task was appropriate for demonstrating the split-attention effect, it may be that the advantage of actively integrating symbolic and pictorial representations would be convincingly revealed only with respect to a more difficult learning task. Therefore, in Experiment 2, we used a much more abstract and complex application domain.

3. Experiment 2

3.1. Method

3.1.1. Design

In the second experiment, we encouraged learners to actively integrate symbolic representations and static versions of dynamic and interactive visualisations in the external environment. Furthermore, we encouraged them to interact with the

dynamic and interactive visualisations in a structured and reflective way. Two factors were varied: (1) the integration of information (presentation of information in a split-source format, presentation of information that was already integrated, active integration of information) and (2) the structuring of interactions (unstructured free exploration versus structured testing of hypotheses).

The three experimental conditions for factor 1 corresponded to the first, second, and fourth condition of Experiment 1: while for two of the conditions (split source or integrated formats), the learners had only to relate the presented information mentally (cf. Chandler & Sweller, 1991), the learners in the third condition had to relate the different representations externally. The two experimental conditions for the second factor differed with respect to the amount of guidance given. While students in the first condition interacted with the dynamic visualisations in a self-guided way, students in the second condition were provided with informative examples of data, which had the potential to lead them to the discovery of important relationships. In addition, the students were encouraged to formulate, test and modify hypotheses in a structured and reflective way. The support measures are explained in more detail below.

3.1.2. *Participants*

Overall, 84 students (65 females and 19 males) in the social sciences at the universities of Freiburg and Tuebingen, aged 20–39 years, were randomly assigned to the six experimental groups. They were paid for their participation. All students had attended courses in introductory and advanced statistics for 1 year at most before the experiment took place but were largely unfamiliar with the visualisations used in this experiment.

3.1.3. *Materials and procedure*

The application domain was various statistics concepts and principles underlying the one-way ANOVA, such as the concepts of error and squared error, the principle of least squares and the method of partitioning the sum of squares.

The learning materials consisted of (1) a printed text, (2) dynamic and interactive visualisations on a computer, and (3) static illustrations with verbal, algebraic, and graphic components on a computer. The visualisations were taken from the interactive learning environment VISUALSTAT¹ (Ploetzner, Bodemer, & Feuerlein, 2001). Learning time was constant under all experimental conditions. The learning procedure comprised six different phases.

3.1.3.1 Phase 1: activation of pre-knowledge. Initially, all students refreshed their knowledge about statistics by means of an instructional text, comprising words and formulas but no pictures (30 min).

3.1.3.2 Phase 2: pre-test. The students took a multiple choice test, made up of four different types of questions: (1) pre-knowledge, (2) retention, (3) verbal understanding and (4) visual understanding. Pre-knowledge questions assessed

¹ <http://www.psychologie.uni-freiburg.de/visualstat/>

general knowledge of the ANOVA. Retention questions were related to factual information directly given in the instructional text. Answering the verbal and visual understanding questions required reasoning and transfer. The pre-test included three questions of each type.

3.1.3.3 Phase 3: introduction to the visualisations. The students received a general introduction to the different visualisations, during which the experimenter gave a short and standardised explanation of the meaning of the diagrams and the possibilities of interaction (5 min).

3.1.3.4 Phase 4: relating representations. Before the learners started to interact with the visualisations, they were provided with static versions of the dynamic and interactive visualisations accompanied by textual and algebraic information from the instructional text (10 min).

In the condition *non-integrated information*, the visualisations, the textual, and the algebraic components were presented in a split-source format (cf. Fig. 2). The relations between components of the different representations were indicated by corresponding numbers.

In the condition *integrated information*, the information was presented in an integrated format (cf. Fig. 2).

In the condition *active integration of information*, the learners were asked to actively relate the textual and algebraic components to the components of the visualisations by drag and drop (cf. Fig. 2). By doing so, the learners themselves constructed an integrated format step by step. After having mapped all components, the students were able to check the correctness of their constructions and, if desired, view the correct solution.

3.1.3.5 Phase 5: interacting with the visualisations. All students interacted with the dynamic visualisations (45 min). In this phase, information about various statistical concepts was presented simultaneously (cf. Fig. 3). The amount of guidance received by the students differed according to experimental condition.

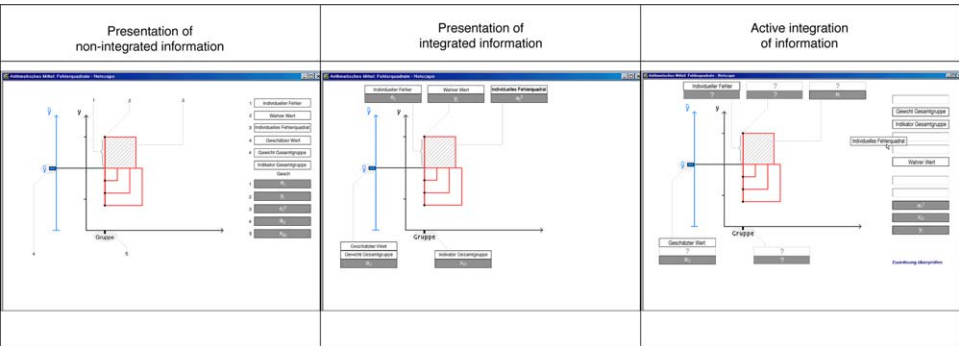


Fig. 2. Variation of the factor integration of information.

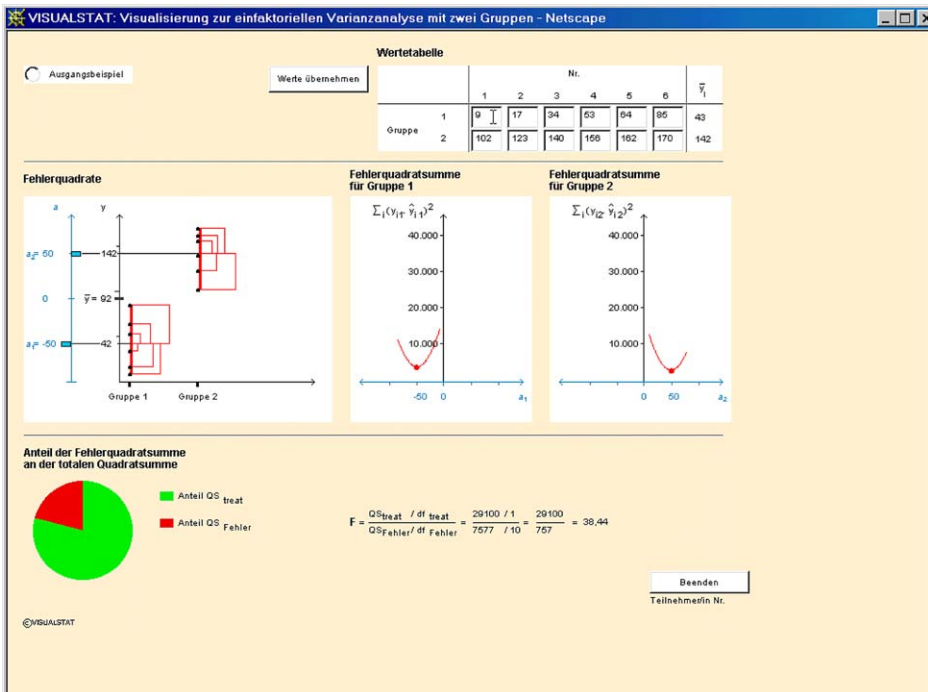


Fig. 3. Dynamic and interactive visualisations of various concepts underlying the one-way ANOVA.

In the *free exploration* condition, learners were able to interact with the visualisations in a self-guided way by creating own data sets and by moving a slide control.

In the *structured testing of hypotheses* condition, learners were provided with four examples of data that had the potential to lead them to discover of important relationships. However, they could not create own data sets. Furthermore, the learners were encouraged to formulate and test hypotheses about the effects of moving the slide control for each presented data set and about the differences between the corresponding graphs of the data sets. In doing so, the learners were guided to focus only on one relevant aspect of a visualisation at a time.

3.1.3.6 Phase 6: post-test. The students worked on an extended version of the pre-test described above. The post-test consisted of eight retention questions, 12 verbal understanding questions, and 12 visual understanding questions.

3.2. Results

In the pre-test, there were no statistically significant differences between the groups. Therefore, in the following, only the results of the post-test are presented.

To make them easier to compare, the test scores were transformed to relative frequencies of correct answers above chance level. Table 3 shows the means and the standard deviations for retention, verbal understanding, and visual understanding for all six experimental conditions. Table 4 shows the results of the multivariate and univariate two-way ANOVA.

On average, learners who actively integrated different representations in the external environment outperformed learners who were provided with split-source formats or pre-integrated representations. Somewhat surprisingly, learners who were provided with pre-integrated representations did not learn significantly more than learners who were provided split-source formats.

Statistically, the structured testing of hypotheses proved to be beneficial with respect to verbal understanding. Concerning retention and visual understanding, those learners in particular who had actively integrated different representations were not able to take additional advantage of the structured testing of hypotheses. Furthermore, there was no statistically significant interaction between the two factors.

3.3. Discussion

The second experiment revealed that learning with dynamic and interactive visualisations can significantly be improved by encouraging learners to actively integrate symbolic and static versions of pictorial representations in the external environment. The results also demonstrate that guided discovery learning can result in a certain degree of learning improvement. However, those students who actively integrated different representations were overall not able to improve their learning outcomes further by means of the structured testing of hypotheses.

While in the first experiment we replicated the finding of John Sweller and his colleagues that an integrated format leads to significantly more learning than a split-source format (e.g., Chandler & Sweller, 1991; Sweller & Chandler, 1994), in the second experiment such an advantageous effect of an integrated format was much less pronounced. The lack of a strong split-attention effect in Experiment 2 may be due to the fact, that even the integrated format still required a considerable amount of visual search. While in Experiment 1 the textual information was presented close to the corresponding pictorial information, in Experiment 2 it was essentially presented outside of the visualisations. Moreover, the learning task in Experiment 2 was far more abstract, and students had to process and relate many more elements simultaneously than in Experiment 1 (cf. Sweller, 1994; Tuovinen & Sweller, 1999). A possible explanation of our results is that while the split-attention effect occurs mostly in the case of a certain amount of difficulty and structural complexity of the learning material (cf. Kalyuga, Ayres, Chandler, & Sweller, 2003), the active integration of information might be most beneficial when the learning material is of particularly high difficulty and complexity, as it was the case in Experiment 2.

Table 3
The means (M) and the standard deviations (SD) for retention and understanding under all six experimental conditions

Integration of information	Retention	Structuring of interactions			Verbal understanding			Visual understanding		
		Self-guided			Self-guided			Self-guided		
		Self-guided	Structured	Overall	Self-guided	Structured	Overall	Self-guided	Structured	Overall
Split-source	M	0.25	0.29	0.27	0.08	0.20	0.14	0.23	0.27	0.25
	SD	0.28	0.28	0.27	0.18	0.15	0.17	0.20	0.14	0.17
Integrated	M	0.26	0.23	0.25	0.14	0.23	0.18	0.20	0.34	0.27
	SD	0.27	0.27	0.26	0.25	0.13	0.20	0.24	0.13	0.20
Actively integrated	M	0.49	0.37	0.43	0.35	0.41	0.38	0.41	0.38	0.40
	SD	0.27	0.31	0.29	0.16	0.21	0.19	0.23	0.28	0.25
Overall	M	0.33	0.30	0.32	0.19	0.28	0.24	0.28	0.33	0.31
	SD	0.29	0.28	0.28	0.23	0.19	0.21	0.24	0.19	0.22

Table 4

The results of the multivariate (Wilks-Lambda) and univariate two-way ANOVA

Source of variance	Dependent variable	df	F
Integration of information	Across retention and understanding	6, 152	4.33**
	Retention	2, 78	3.58*
	Verbal understanding	2, 78	13.73**
	Visual understanding	2, 78	4.24*
Structuring of interactions	Across retention and understanding	3, 76	2.31
	Retention	1, 78	0.35
	Verbal understanding	1, 78	4.80*
	Visual understanding	1, 78	1.15
Integration of information × structuring of interactions	Across retention and understanding	6, 152	0.66
	Retention	2, 78	0.59
	Verbal understanding	2, 78	0.15
	Visual understanding	2, 78	1.18

* $p < 0.05$.** $p < 0.01$.

4. General discussion

Dynamic and interactive visualisations have the potential to improve computer-based learning in various ways, but—especially if they are presented in combination with other kinds of external representations—their mental processing can also be very demanding and so easily overburden the learners' cognitive capacities. As a consequence, learners frequently limit their attention to the most perceptually salient but not necessarily the most thematically relevant features of the visualisations, do not interact with them in a structured and goal-oriented way, and do not systematically relate the different sources of information to each other in order to construct a coherent mental representation.

While the past focus of research on cognitive load during learning was mainly on how extraneous cognitive load can be reduced, very recently the focus in various studies has changed to how germane cognitive load can be increased (e.g., Kirschner, 2002; Sweller, 1988; van Merriënboer et al., 2002). In our research, we took advantage of cognitive load theory (Sweller, 1994) and structure mapping theory (Gentner, 1983; Gentner & Markman, 1997) to design learning experiences such that they both decrease extraneous cognitive load and increase germane cognitive load. In order to support the mental integration of different sources of information, we encouraged learners to map interactively components of familiar representations to components of unfamiliar representations in the external environment.

While Mayer (2001); Robins & Mayer, 1993) assumes that multimedia learning relies exclusively on mental processes and that actions in the external environment do not necessarily contribute to multimedia learning, the results of our experiments demonstrate that it can be important to consider external activities in instructional design. Of course in the final analysis, multimedia learning is always the result of applying specific mental processes. Furthermore, many students may have learned

to apply spontaneously appropriate mental processes when faced with multimedia learning materials. Many students, however, may have not. How then can these students be supported in processing the learning materials adequately? We propose that tasks in the external environment that can be accomplished successfully only when the relevant mental processes take place can provide one such support measure. If such a task is taken seriously by the students, the attempt to fulfil the task by means of external actions at the same time induces and supports the execution of the mental processes in question.

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